SpectonCR on EC2 + S3

Deployment guide and Amazon ECR cost comparison

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1. Running SpectonCR on EC2 with S3 blob storage

Yes — SpectonCR ships **native S3 storage support**, so EC2 + S3 is a first-class deployment. The image blobs and manifests live in S3; only a tiny amount of metadata is kept in memory or on the local disk of the EC2 instance.

> **TL;DR:** Run the SpectonCR container on a small Graviton EC2 instance, point its `[storage]` backend at an S3 bucket in the same region, attach an IAM instance profile, and add a VPC S3 Gateway Endpoint. That's the whole architecture.

2. Wiring up S3 as the storage backend

Config file (`config/spectoncr.example.toml`, lines 129–161)

```
[storage]
backend = "s3"
root    = "my-spectoncr-bucket"     # S3 bucket name
region  = "eu-west-2"
# endpoint = unset on real AWS S3
# Leave access_key / secret_key UNSET — use the EC2 IAM instance profile instead
```

Or via environment variables (no file edits needed)

```
SPECTONCR_STORAGE__BACKEND=s3
SPECTONCR_STORAGE__ROOT=my-spectoncr-bucket
SPECTONCR_STORAGE__REGION=eu-west-2
```

Run the container

```
docker run -d --name spectoncr -p 5000:5000 -p 5001:5001 \
  -e SPECTONCR_STORAGE__BACKEND=s3 \
  -e SPECTONCR_STORAGE__ROOT=my-spectoncr-bucket \
  -e SPECTONCR_STORAGE__REGION=eu-west-2 \
  specton/spectoncr:latest
```

> **IAM permissions** for the EC2 instance role — minimum set on the bucket: `s3:GetObject`, `s3:PutObject`, `s3:DeleteObject`, `s3:ListBucket`, `s3:AbortMultipartUpload`. No static access keys are required.

3. Cost-saving suggestions (biggest wins first)

1. **S3 Gateway Endpoint** in your VPC (free). Without it, EC2 → S3 traffic hops through NAT and you pay ~**$0.045/GB**. The gateway endpoint makes it free.
2. **Put EC2 + S3 bucket in the same region.** Cross-region replication and egress charges add up fast for a registry.
3. **Graviton EC2** (`t4g.small` / `c7g.medium`). SpectonCR publishes `linux/arm64` images. ~20% cheaper than x86 equivalents.
4. **Right-size + Spot for non-prod.** A `t4g.small` (2 vCPU, 2 GB) handles a small team easily since blobs are streamed straight to/from S3. For prod, a Compute Savings Plan on a `t4g.medium` is the sweet spot.
5. **S3 lifecycle rules:**
   - Transition blobs to **S3 Intelligent-Tiering** (auto-moves cold blobs to IA / Glacier Instant; no retrieval fee for IA).
   - Or hard rule: blobs untouched 30 days → Standard-IA, 90 days → Glacier Instant Retrieval.
   - Set an **AbortIncompleteMultipartUpload** rule (7 days) — registry pushes leave orphans on failure.
6. **Enable SpectonCR pull-through cache** (`[mirror]` section). Once a Docker Hub image is fetched, subsequent pulls hit S3 instead of Docker Hub — kills Docker Hub rate-limit pain and saves egress for downstream nodes.
7. **CloudFront in front of `/v2/`** if you have lots of external pulls. Egress from CloudFront is cheaper than direct S3/EC2 egress, and it caches blobs at the edge.
8. **Single-AZ EBS + no ALB** for a dev registry. Skip the ALB (~$16/mo idle) and put the EC2 directly behind an Elastic IP + Route 53, or run Caddy/Nginx on the same box for TLS termination.
9. **Run garbage collection regularly** (untagged manifests + orphaned blobs) to keep S3 storage growth bounded.
10. **Turn off S3 versioning** unless you really need it — every overwrite otherwise doubles storage.

> **Rough monthly estimate** for a small team registry: `t4g.small` (~$12) + 50 GB S3 Standard (~$1.15) + minimal egress through the VPC endpoint = **under $20/month all-in**.

4. Amazon ECR cost for 300 GB of images

ECR's pricing model is dead simple — storage + data transfer out.

Storage

- **$0.10 per GB-month** (ECR Private, all regions)
- 300 GB × $0.10 = **$30.00 / month** just for storage

Data transfer (the part that bites people)

Path	Price	Notes
Push to ECR	Free	Always
Pull → EC2 in same region & AZ	Free	Use this
Pull → EC2 in same region, different AZ	$0.01/GB	
Pull → EC2 in different region	$0.02/GB	
Pull → public internet	$0.09/GB (first 10 TB)	Hurts

Realistic scenarios at 300 GB stored

Usage profile	Monthly cost
Just sitting there, no pulls	**$30**
Same-region/AZ pulls only (typical EKS / ECS)	**$30**
500 GB/mo pulled cross-AZ	**$30 + $5 = $35**
1 TB/mo pulled to the internet (dev laptops, CI on GitHub)	**$30 + $90 = $120**

5. ECR vs SpectonCR + S3 side-by-side

Same workload on EC2 + S3 (SpectonCR)

Item	Cost
S3 Standard storage, 300 GB	300 × $0.023 = **$6.90**
With Intelligent-Tiering after 30 days idle	drops to ~$3–4 for cold blobs
`t4g.small` EC2 (on-demand)	~$12.30
30 GB gp3 EBS	~$2.40
S3 → EC2 via Gateway Endpoint	**Free**
Total	**~$22 / month**

> **Storage alone:** ECR **$30** vs S3 **$7** — S3 is ~4× cheaper per GB. Add ~$15 for the EC2 box and you still come out under ECR, *and* you skip ECR's cross-AZ data transfer charges.

Where ECR is still worth it

- You don't want to operate anything — ECR is fully managed, HA, IAM-native, integrated with EKS/ECS/Lambda.
- Image scanning (basic is free, enhanced via Inspector is $0.09 / image scan).
- Cross-region replication is built-in (but billed as storage in each region).

6. Quick optimizations if you stick with ECR

1. **Lifecycle policy** — auto-delete untagged images after N days; keep only last 10 tags per repo. This alone usually halves storage.
2. **Pull cache** in your cluster (e.g. Spegel, Harbor proxy cache) to avoid re-pulling the same image to every node.
3. **VPC endpoint for ECR** (`com.amazonaws.<region>.ecr.dkr` + `.ecr.api` + S3 gateway) — kills NAT data charges when nodes are in private subnets.
4. Keep workloads + ECR in the **same region**, and ideally the same AZ for the puller.

7. Bottom line

> At 300 GB:
>
> - **ECR** ≈ **$30/mo** storage before any pulls, and easily $50–$120+ with realistic data transfer.
> - **SpectonCR on EC2 + S3** lands at **~$15–$22/mo all-in**, and the per-GB curve stays gentle as your image set grows.